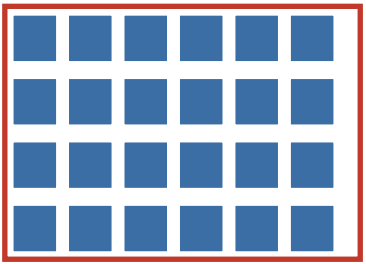
Dense 70B

ALL active



all 140 GB resident & active

every token activates all 70B params

KV grows linearly with context

dense 12 GB vs MoE 0.6 GB

8-bit KV set by shape, not routing:
80L/8192H MHA vs 32L/4096H GQA
(8 KV heads × 128), at 9.2K tokens

MoE (8 experts)

top-2 active

inactive experts ≈ 68 GB (6 × 11.4)

active experts ≈ 23 GB resident (2 × 11.4)

MoE total resident ≈ 94 GB
= 91 GB experts + 3 GB always-on
per token: 11.4B top-2 + 1.4B always-on
= 12.8B → 25.6 GB

KV grows linearly with context

dense 12 GB vs MoE 0.6 GB

8-bit KV set by shape, not routing:
80L/8192H MHA vs 32L/4096H GQA
(8 KV heads × 128), at 9.2K tokens